

# Supplementary Material: Environment Sensitivity and Reproducibility of Post-BA SfM Benchmarks

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## A. What this document supports

This supplementary substantiates four specific statements in the main paper; a reviewer following any of its pointers can go directly to the listed section.

- Protocol, “Track provenance”: “ $\sim 0.209$  is the reproducible level [for the released RESfM checkpoint] ... unchanged (0.217) under the authors’ exact pinned software stack” (Secs. B–D).
- Protocol: “a stack-replication study (supplementary) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism” (Sec. E).
- Table 2, † footnote: “TTT is run-to-run nondeterministic” (Sec. E).
- Reading any post-BA mean in this literature: differences  $\lesssim 0.05$  are within environment noise (Sec. F).

In addition, Sec. G tabulates the per-scene translation, rotation, and registered-camera ( $N_r$ ) results behind the main paper’s dataset-level numbers, with dataset-level rotation and registration summaries in Table 11; Sec. H gives the complete training and evaluation configuration, Sec. I provides the per-scene label-degradation breakdown behind the label-ceiling analysis, and Sec. J provides extended versions of the main paper’s Tables 2, 5, 6, and 7 (adding rotation and  $N_r$ ) intended to replace them in the camera-ready.

## B. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation 1.89° vs. 1.29°). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry a practical environment-noise floor of  $\mathcal{O}(0.05)$ , produced by chaotic basin selection in robust bundle adjustment.

Table 8. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number.  $N_r$  is the mean number of registered cameras per scene; the published row averages the paper’s per-scene Table 1 values.

| Condition                           | Trans | Rot (°) | $N_r$ |
|-------------------------------------|-------|---------|-------|
| Published (paper Table 1)           | 0.169 | 1.29    | 298   |
| Our env (pip/OpenBLAS), 5-seed mean | 0.203 | 1.82    | 304   |
| Our env, seed 20                    | 0.209 | 1.89    | 304   |
| Authors’ exact stack (conda/MKL)    | 0.217 | 1.92    | 304   |

## C. Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by  $\pm 0.004$  and the post-BA mean by  $\pm 0.01$ ; negligible.

Network forward pass. The pre-BA output (translation/rotation  $\sim 1.73/11.9^\circ$ ) is essentially seed- and environment-stable; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives translation/rotation errors of 0.217/1.92°, statistically indistinguishable from our environment’s 0.209/1.89°, and not the published 0.169/1.29°. Registration is equally stable: all three of our conditions register 304 cameras per scene on average (Table 8), slightly more than the published 298, so the higher mean error is not an artifact of evaluating fewer cameras.

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| 107 | The attribution is therefore not the software stack but                   | in rotation ( $36^\circ$ ) and the high-contamination order-         | 157 |
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Table 9. Per-scene 1DSfM results (layout follows RESfM’s Table 1). For each scene we show the number of input images ( $N_c$ ) and the fraction of outlier observations; for each model, the number of registered cameras ( $N_r$ ) and mean rotation ( $e_R$ , degrees) and translation ( $e_t$ ) errors, post-BA, seed 20. Scenes are ordered by outlier fraction. Winning results per scene are marked in bold and underlined.

| Scene               | $N_c$ | %out | madweight |                     |                    | weight |                    |                     | weight+TTT |                     |                    | RESfM |                    |                    |
|---------------------|-------|------|-----------|---------------------|--------------------|--------|--------------------|---------------------|------------|---------------------|--------------------|-------|--------------------|--------------------|
|                     |       |      | $N_r$     | $e_R$               | $e_t$              | $N_r$  | $e_R$              | $e_t$               | $N_r$      | $e_R$               | $e_t$              | $N_r$ | $e_R$              | $e_t$              |
| Ellis_Island        | 227   | 56%  | 180       | <b><u>17.16</u></b> | 17.59              | 171    | 17.37              | <b><u>15.90</u></b> | 177        | 17.74               | 17.11              | 192   | 18.15              | 16.02              |
| NYC_Library         | 332   | 54%  | 282       | 2.32                | 2.94               | 261    | <b><u>1.16</u></b> | <b><u>1.19</u></b>  | 279        | 10.30               | 3.39               | 285   | 12.99              | 6.28               |
| Vienna_Cathedral    | 836   | 51%  | 537       | 7.42                | 7.30               | 484    | <b><u>0.46</u></b> | 22.71               | 651        | 26.80               | 33.57              | 492   | 1.09               | <b><u>2.29</u></b> |
| Notre_Dame          | 553   | 48%  | 507       | 3.33                | 0.68               | 491    | 1.00               | 0.34                | 485        | <b><u>0.83</u></b>  | <b><u>0.33</u></b> | 493   | 1.08               | 0.49               |
| Yorkminster         | 437   | 45%  | 299       | 8.05                | <b><u>5.79</u></b> | 371    | 14.95              | 30.87               | 380        | 12.20               | 10.79              | 309   | <b><u>6.88</u></b> | 8.77               |
| Piazza_del_Popolo   | 338   | 42%  | 218       | <b><u>0.43</u></b>  | <b><u>0.94</u></b> | 265    | 19.54              | 10.66               | 254        | 20.46               | 17.05              | 255   | 3.15               | 2.49               |
| Tower_of_London     | 472   | 42%  | 317       | 35.61               | 76.34              | 337    | 23.30              | <b><u>28.29</u></b> | 314        | <b><u>17.91</u></b> | 72.61              | 299   | 49.66              | 67.95              |
| Montreal_Notre_Dame | 450   | 36%  | 311       | 0.59                | 0.94               | 325    | <b><u>0.55</u></b> | 11.35               | 320        | 0.81                | 1.24               | 400   | 0.59               | <b><u>0.93</u></b> |
| Alamo               | 577   | 29%  | 467       | 2.45                | 1.77               | 507    | 2.74               | 2.18                | 497        | 2.49                | <b><u>1.67</u></b> | 502   | <b><u>2.42</u></b> | 1.73               |
| Madrid_Metropolis   | 341   | 29%  | 249       | 5.72                | 9.63               | 301    | 2.84               | 4.37                | 295        | 2.35                | 3.52               | 287   | <b><u>1.82</u></b> | <b><u>3.38</u></b> |

## H. Implementation details

Table 12 lists the full training and evaluation configuration. Both methods share every setting except the base learning rate ( $10^{-4}$  self-supervised,  $10^{-3}$  supervised, each selected on its own training objective) and the supervision source (COLMAP-derived labels vs. the adaptive percentile pseudo-labels).

Table 10. Per-scene MegaDepth results (layout follows RESfM’s Table 1). Columns as in Table 9; seed 20. Above the middle rule are Group 1 scenes (< 1000 images); below are Group 2 scenes (> 1000 images, subsampled to 300 for testing), each group ordered by outlier fraction. Winning results per scene are marked in bold and underlined. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. E).

| Scene | $N_c$ | %out | madweight |                    |                    | weight |                    |                    | weight+TTT |                    |                    | RESfM |                    |                    |
|-------|-------|------|-----------|--------------------|--------------------|--------|--------------------|--------------------|------------|--------------------|--------------------|-------|--------------------|--------------------|
|       |       |      | $N_r$     | $e_R$              | $e_t$              | $N_r$  | $e_R$              | $e_t$              | $N_r$      | $e_R$              | $e_t$              | $N_r$ | $e_R$              | $e_t$              |
| 0238  | 522   | 45%  | 437       | 19.02              | 1.41               | 137    | 8.23               | 0.62               | 141        | 8.37               | 0.63               | 355   | <b><u>1.40</u></b> | <b><u>0.12</u></b> |
| 0060  | 528   | 42%  | 469       | 0.29               | 0.03               | 476    | 0.20               | 0.02               | 452        | 4.13               | 0.37               | 494   | <b><u>0.04</u></b> | <b><u>0.01</u></b> |
| 0197  | 870   | 41%  | 470       | 7.22               | 0.89               | 341    | 3.54               | 0.38               | 422        | 6.94               | 1.11               | 739   | <b><u>0.97</u></b> | <b><u>0.11</u></b> |
| 0094  | 763   | 40%  | 493       | 2.72               | 0.45               | 505    | 1.44               | <b><u>0.16</u></b> | 432        | <b><u>1.34</u></b> | 0.17               | 572   | 5.33               | 0.88               |
| 0265  | 571   | 39%  | 457       | 17.62              | 3.26               | 477    | 5.78               | 2.34               | 547        | <b><u>0.60</u></b> | <b><u>0.19</u></b> | 424   | 16.27              | 3.09               |
| 0083  | 635   | 31%  | 342       | 4.34               | 0.23               | 511    | 17.70              | 1.56               | 317        | 3.20               | 0.20               | 582   | <b><u>1.40</u></b> | <b><u>0.10</u></b> |
| 0076  | 558   | 30%  | 423       | 1.40               | 0.38               | 519    | 0.61               | 0.23               | 454        | 0.74               | 0.22               | 499   | <b><u>0.25</u></b> | <b><u>0.02</u></b> |
| 0185  | 368   | 30%  | 347       | 0.21               | 0.01               | 335    | 0.24               | 0.04               | 321        | 0.88               | 0.09               | 352   | <b><u>0.07</u></b> | <b><u>0.01</u></b> |
| 0048  | 512   | 24%  | 475       | 1.93               | 0.14               | 467    | <b><u>0.35</u></b> | <b><u>0.02</u></b> | 450        | 0.43               | 0.03               | 489   | 1.85               | 0.11               |
| 0024  | 356   | 23%  | 299       | 1.27               | 0.18               | 291    | <b><u>1.13</u></b> | <b><u>0.17</u></b> | 289        | 4.78               | 1.13               | 334   | 3.67               | 1.09               |
| 0223  | 214   | 17%  | 205       | 2.66               | 0.24               | 198    | 1.69               | 0.23               | 199        | 3.06               | 0.30               | 204   | <b><u>0.45</u></b> | <b><u>0.08</u></b> |
| 5016  | 28    | 17%  | 23        | 0.27               | 0.04               | 25     | 0.08               | 0.02               | 28         | <b><u>0.07</u></b> | 0.02               | 28    | 0.08               | <b><u>0.01</u></b> |
| 0046  | 440   | 15%  | 422       | <b><u>2.88</u></b> | <b><u>0.22</u></b> | 421    | 4.11               | 0.30               | 419        | 3.88               | 0.25               | 410   | 14.64              | 0.93               |
| 0099  | 299   | 47%  | 224       | <b><u>0.81</u></b> | 0.15               | 216    | 2.77               | 3.52               | 212        | 1.58               | <b><u>0.13</u></b> | 252   | 2.18               | 0.39               |
| 1001  | 285   | 44%  | 280       | 1.58               | 1.22               | 278    | 1.49               | 0.73               | 279        | <b><u>1.22</u></b> | <b><u>0.49</u></b> | 273   | 1.23               | 0.50               |
| 0231  | 296   | 42%  | 274       | 1.63               | 0.15               | 257    | 0.64               | 0.07               | 264        | 1.49               | 0.24               | 265   | <b><u>0.17</u></b> | <b><u>0.02</u></b> |
| 0411  | 299   | 30%  | 281       | 0.22               | 0.03               | 245    | 2.99               | 0.32               | 250        | 4.12               | 0.46               | 275   | <b><u>0.11</u></b> | <b><u>0.02</u></b> |
| 0377  | 295   | 28%  | 200       | 0.62               | 0.05               | 222    | 2.76               | 0.26               | 243        | <b><u>0.56</u></b> | <b><u>0.03</u></b> | 223   | 0.82               | 0.06               |
| 0102  | 299   | 26%  | 289       | 0.15               | 0.02               | 289    | <b><u>0.07</u></b> | <b><u>0.01</u></b> | 291        | 0.08               | 0.01               | 287   | 0.10               | 0.02               |
| 0147  | 298   | 25%  | 261       | 31.69              | 2.99               | 235    | 15.28              | 1.21               | 225        | 15.27              | <b><u>1.20</u></b> | 247   | <b><u>4.61</u></b> | 3.22               |
| 0148  | 287   | 25%  | 220       | 2.60               | 0.33               | 201    | 2.11               | 0.11               | 196        | <b><u>0.62</u></b> | 0.04               | 202   | 0.94               | <b><u>0.04</u></b> |
| 0446  | 298   | 22%  | 292       | <b><u>0.40</u></b> | <b><u>0.03</u></b> | 285    | 1.20               | 0.19               | 292        | 1.04               | 0.19               | 293   | 0.71               | 0.05               |
| 0022  | 297   | 21%  | 275       | 0.14               | 0.02               | 272    | 0.22               | 0.03               | 267        | 0.33               | 0.04               | 276   | <b><u>0.11</u></b> | <b><u>0.02</u></b> |
| 0327  | 298   | 21%  | 282       | 1.97               | 0.09               | 285    | 0.22               | 0.02               | 284        | 0.21               | 0.02               | 277   | <b><u>0.09</u></b> | <b><u>0.01</u></b> |
| 0015  | 284   | 21%  | 196       | 2.64               | 0.30               | 196    | 3.87               | 0.56               | 211        | 3.90               | 0.65               | 232   | <b><u>1.31</u></b> | <b><u>0.30</u></b> |
| 0455  | 298   | 20%  | 283       | <b><u>0.20</u></b> | <b><u>0.01</u></b> | 285    | 0.21               | 0.01               | 288        | 0.46               | 1.60               | 293   | 0.85               | 0.09               |
| 0496  | 297   | 19%  | 279       | <b><u>0.27</u></b> | 0.04               | 285    | 0.50               | <b><u>0.04</u></b> | 287        | 1.28               | 0.09               | 283   | 0.46               | 0.07               |
| 1589  | 299   | 17%  | 265       | 0.96               | 0.06               | 262    | 0.28               | 0.05               | 296        | 1.91               | 0.30               | 291   | <b><u>0.06</u></b> | <b><u>0.01</u></b> |
| 0012  | 299   | 16%  | 281       | <b><u>0.23</u></b> | <b><u>0.02</u></b> | 247    | 9.51               | 0.70               | 262        | 2.76               | 0.22               | 288   | 5.53               | 0.34               |
| 0104  | 284   | 16%  | 189       | 0.32               | 0.03               | 193    | 2.28               | 0.18               | 194        | 1.44               | 0.07               | 199   | <b><u>0.30</u></b> | <b><u>0.02</u></b> |
| 0019  | 299   | 15%  | 232       | 3.22               | 0.17               | 253    | 6.55               | 0.28               | 264        | 3.24               | 0.13               | 252   | <b><u>0.05</u></b> | <b><u>0.01</u></b> |
| 0063  | 293   | 14%  | 258       | 0.63               | 0.08               | 268    | 0.72               | 0.07               | 267        | 0.65               | <b><u>0.06</u></b> | 266   | <b><u>0.44</u></b> | 0.07               |
| 0130  | 285   | 14%  | 250       | 10.17              | 1.60               | 184    | 1.87               | 0.16               | 185        | <b><u>0.33</u></b> | <b><u>0.01</u></b> | 195   | 1.03               | 0.07               |
| 0080  | 284   | 13%  | 132       | 1.10               | <b><u>0.05</u></b> | 148    | 2.48               | 0.20               | 147        | 0.70               | 0.07               | 141   | <b><u>0.66</u></b> | 0.10               |
| 0240  | 298   | 12%  | 275       | <b><u>2.23</u></b> | <b><u>0.25</u></b> | 275    | 4.64               | 0.34               | 272        | 5.32               | 0.46               | 282   | 3.02               | 0.35               |
| 0007  | 290   | 12%  | 106       | <b><u>0.07</u></b> | <b><u>0.01</u></b> | 269    | 1.86               | 0.09               | 261        | 3.28               | 0.14               | 104   | 0.99               | 0.04               |

Table 11. Dataset-level translation, rotation, and registered cameras (post-BA, seed 20) for the four main-comparison arms. Per-scene values are camera means;  $N_r$  is the fraction of ground-truth cameras surviving robust BA and the 5-pixel reprojection quality mask, summed over the dataset’s scenes.

|                                                        | madweight   | weight      | weight+TTT  | RESfM       |
|--------------------------------------------------------|-------------|-------------|-------------|-------------|
| Translation error (mean / median over scenes)          |             |             |             |             |
| MegaDepth                                              | 0.422/0.145 | 0.423/0.182 | 0.316/0.186 | 0.344/0.07  |
| 1DSfM                                                  | 12.4/4.37   | 12.8/11.0   | 16.1/7.16   | 11.0/2.93   |
| 1DSfM-hard                                             | 12.2/8.83   | 21.9/13.1   | 26.4/17.7   | 16.2/13.1   |
| Strecha                                                | 2.5/0.97    | 2/0.026     | 1.98/0.005  | 0.006/0.005 |
| BlendedMVS                                             | 0.146/0.004 | 0.1/0.036   | 0.136/0.011 | 0.379/0.348 |
| Olsson                                                 | 2.57/1.15   | 2.65/1.22   | 7.74/0.18   | 8.47/0.17   |
| Rotation error ( $^\circ$ , mean / median over scenes) |             |             |             |             |
| MegaDepth                                              | 3.49/1.33   | 3.04/1.77   | 2.51/1.39   | 2.01/0.84   |
| 1DSfM                                                  | 8.31/4.53   | 8.39/2.79   | 11.19/11.25 | 9.79/2.79   |
| 1DSfM-hard                                             | 14.40/7.39  | 25.58/17.70 | 31.97/12.83 | 22.70/15.86 |
| Strecha                                                | 14.62/8.43  | 7.48/0.17   | 7.41/0.02   | 0.02/0.01   |
| BlendedMVS                                             | 9.60/0.38   | 6.65/3.06   | 7.91/0.55   | 36.10/29.55 |
| Olsson                                                 | 13.57/4.37  | 13.23/5.41  | 10.49/2.27  | 12.34/3.35  |
| Registered cameras $N_r$                               |             |             |             |             |
| MegaDepth                                              | 80%         | 79%         | 78%         | 85%         |
| 1DSfM                                                  | 74%         | 77%         | 80%         | 77%         |
| 1DSfM-hard                                             | 38%         | 39%         | 43%         | 37%         |
| Strecha                                                | 91%         | 96%         | 98%         | 100%        |
| BlendedMVS                                             | 95%         | 88%         | 95%         | 90%         |
| Olsson                                                 | 83%         | 90%         | 90%         | 88%         |

Table 12. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

| Setting                                | Value                                                                    |
|----------------------------------------|--------------------------------------------------------------------------|
| Training (both methods)                |                                                                          |
| Epochs                                 | 20,000 (one epoch = one pass over all scenes)                            |
| Batch                                  | 4 scenes                                                                 |
| LR schedule                            | milestones 10k, 11k, $\dots$ , 20k, $\gamma = 0.9$                       |
| Base LR                                | $10^{-4}$ (SS-RESfM) / $10^{-3}$ (RESfM)                                 |
| Loss weights                           | $\alpha_{\text{reproj}} = 1$ , $\beta_{\text{cls}} = 1$                  |
| Infinity-point margin                  | $10^{-4}$                                                                |
| Checkpoint selection                   | best validation error, evaluated every 500 epochs                        |
| Seed                                   | 20 (21–22 / 21–24 for the seed studies)                                  |
| Adaptive pseudo-labels (SS-RESfM only) |                                                                          |
| Inlier / outlier percentile            | 20 / 80 (of per-observation reproj. error)                               |
| Threshold separation                   | scale-aware floor, $0.25 \times$ median error                            |
| Warm-up                                | 10 epochs before pseudo-labeling starts                                  |
| Evaluation                             |                                                                          |
| Removal threshold $\tau$               | 0.6 (0.8 for low-contamination single-camera sets, following RESfM)      |
| Test-time fine-tuning                  | 1,001 iterations, LR $5 \times 10^{-3}$ , evaluated every 250 iterations |
| Bundle adjustment                      | pyceres/pycolmap robust BA, as RESfM                                     |

## I. Per-scene label degradation

Table 13 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below  $\sim 10\%$  contamination coverage is 0.92–1.00 and recall is 0.66–0.97 (mostly above 0.8), while above  $\sim 15\%$  contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

Table 13. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Recall collapses above  $\sim 15\%$  contamination.

| Dataset    | Scene             | Contam. (%) | Coverage | Prec. | Recall |
|------------|-------------------|-------------|----------|-------|--------|
| BlendedMVS | 58c4bb4f4a...     | 0.5         | 1.00     | 0.86  | 0.94   |
| Strecha    | Herz-Jesu-P8      | 0.7         | 1.00     | 0.51  | 0.66   |
| Strecha    | fountain-P11      | 0.7         | 1.00     | 0.69  | 0.76   |
| Strecha    | Herz-Jesu-P25     | 2.6         | 0.99     | 0.72  | 0.82   |
| Strecha    | entry-P10         | 2.7         | 0.99     | 0.88  | 0.88   |
| BlendedMVS | 5be883a4f9...     | 4.0         | 0.98     | 0.90  | 0.84   |
| BlendedMVS | 5b6e716d67...     | 6.1         | 0.92     | 0.89  | 0.92   |
| BlendedMVS | 5bfd0f32ec...     | 7.1         | 0.97     | 0.97  | 0.97   |
| BlendedMVS | 5b950c7160...     | 9.2         | 0.96     | 0.92  | 0.94   |
| MegaDepth  | 0007              | 11.2        | 0.73     | 0.84  | 0.47   |
| 1DSfM      | Alamo             | 26.6        | 0.76     | 0.81  | 0.25   |
| 1DSfM      | Madrid_Metropolis | 28.8        | 0.67     | 0.85  | 0.28   |
| MegaDepth  | 0041              | 31.4        | 0.49     | 0.80  | 0.42   |
| MegaDepth  | 0060              | 39.0        | 0.42     | 0.53  | 0.45   |
| 1DSfM      | Piazza_del_Popolo | 41.9        | 0.76     | 0.90  | 0.30   |
| MegaDepth  | 0047              | 54.9        | 0.37     | 0.83  | 0.50   |
| 1DSfM      | Ellis_Island      | 57.9        | 0.79     | 0.89  | 0.37   |
| 1DSfM-hard | Gendarmenmarkt    | 59.8        | 0.50     | 0.89  | 0.29   |

## J. Extended main-paper tables (camera-ready)

Tables 14–17 extend the main paper’s Tables 2, 5, 6, and 7 with rotation error and registered-camera counts alongside translation, following RESfM’s metric emphasis (translation first, then rotation, then  $N_r$ ). All values come from the same seed-20 evaluations behind the main paper (rotation panels and  $N_r$  at seed 20; translation panels repeat the main-paper cells including their seed bands). In the camera-ready we will replace main-paper Tables 2, 5, 6, and 7 with these extended versions. Reading the additions: rotation broadly agrees with translation (the supervised BlendedMVS failure is even larger in rotation,  $36^\circ$ ; the high-contamination ordering is preserved), though the best arm can differ per metric (on MegaDepth, weight+TTT leads translation while the supervised baseline leads rotation). Registration is contamination-dependent: the supervised baseline registers the most cameras in distribution (85%) and on clean Strecha (100%), while under high contamination the label-free arms retain more (38–43% vs. 37% on 1DSfM-hard). In the ablation (Table 16) rotation and registration are statistic-insensitive ( $7.6$ – $9.0^\circ$ ,  $71$ – $74\%$ ): the statistic matters through the translation medians. The selection oracle (Table 17) improves rotation as well as translation (MegaDepth  $1.22^\circ$  vs. the supervised  $2.01^\circ$ ) but registers fewer 1DSfM cameras (66%): selecting by translation error can sacrifice registration.

Two notions of registration, and an error-vs-coverage trade-off. Two registration notions must be distinguished. Main-paper Table 4’s “Registered” column reports pose coverage: every learned arm outputs a pose for every camera (“all”), whereas COLMAP genuinely fails to register 36% of the cameras.  $N_r$  here is stricter: the cameras surviving robust BA and the 5-pixel reprojection quality mask, i.e. the set over which the reported errors are computed. Under this stricter notion the plain-remove arms attain their low errors over a reduced registered set: on 1DSfM plain remove registers 59% vs. the supervised baseline’s 77%, and on 1DSfM-hard 28% vs. 37% (remove+TTT: 59%/32%), consistent with aggressive pruning disconnecting weakly-supported cameras (main-paper limitation 5). The camera-ready will state this caveat alongside the plain-remove claims (“over the cameras it retains”). madweight, by contrast, is registration-matched to the supervised baseline (74%/38% vs. 77%/37%),

Table 14. Extended main comparison (extends main-paper Table 2; RESfM-aligned schedule, seed 20). Translation panel repeats the main table (mean / median;  $^\dagger/^\ddagger$  as there); rotation (mean / median,  $^\circ$ ) and  $N_r$  panels at seed 20. Best per column in bold.

| Arm                                                   | MegaDepth (25.4)                | 1DSfM (43.3)      | 1DSfM-hard (60)                                   | Strecha (1.7)      | BMVS (3.1)          | Olsson (0.5)                                     |
|-------------------------------------------------------|---------------------------------|-------------------|---------------------------------------------------|--------------------|---------------------|--------------------------------------------------|
| Translation error (mean / median), the primary metric |                                 |                   |                                                   |                    |                     |                                                  |
| madweight                                             | 0.422/0.145                     | 12.39/4.37        | 17.0 $\pm$ 4.4/ <b>11.7<math>^\ddagger</math></b> | 2.50/0.97          | 0.146/ <b>0.004</b> | <b>2.8 <math>\pm</math> 0.3</b> /1.2 $^\ddagger$ |
| weight                                                | 0.423/0.182                     | 12.79/11.01       | 21.9 $\pm$ 3.4/12.7 $^\ddagger$                   | 2.00/0.026         | <b>0.100</b> /0.036 | 2.8 $\pm$ 0.4/1.1 $^\ddagger$                    |
| weight+TTT                                            | 0.39 $\pm$ 0.07/0.14 $^\dagger$ | 16.13/7.16        | 21.9 $\pm$ 3.9/14.5 $^\ddagger$                   | 1.98/ <b>0.005</b> | 0.136/0.011         | 8.0 $\pm$ 0.4/0.4 $^\ddagger$                    |
| RESfM-scratch (supervised)                            | <b>0.344/0.070</b>              | <b>11.03/2.93</b> | <b>16.22</b> /13.08                               | <b>0.006/0.005</b> | 0.379/0.348         | 8.47/ <b>0.17</b>                                |
| RESfM (released)                                      | 0.209                           | 10.71             | —                                                 | 0.197              | 0.330               | —                                                |
| RESfM (paper)                                         | 0.169                           | —                 | —                                                 | —                  | —                   | —                                                |
| Rotation error ( $^\circ$ , mean / median)            |                                 |                   |                                                   |                    |                     |                                                  |
| madweight                                             | 3.49/1.33                       | <b>8.31</b> /4.53 | <b>14.40/7.39</b>                                 | 14.62/8.43         | 9.60/0.38           | 13.57/4.37                                       |
| weight                                                | 3.04/1.77                       | 8.39/ <b>2.79</b> | 25.58/17.70                                       | 7.48/0.17          | <b>6.65</b> /3.06   | 13.23/5.41                                       |
| weight+TTT                                            | 2.51/1.39                       | 11.19/11.25       | 31.97/12.83                                       | 7.41/0.02          | 7.91/ <b>0.55</b>   | <b>10.49/2.27</b>                                |
| RESfM-scratch (supervised)                            | <b>2.01/0.84</b>                | 9.79/ <b>2.79</b> | 22.70/15.86                                       | <b>0.02/0.01</b>   | 36.10/29.55         | 12.34/3.35                                       |
| Registered cameras $N_r$                              |                                 |                   |                                                   |                    |                     |                                                  |
| madweight                                             | 80%                             | 74%               | 38%                                               | 91%                | <b>95%</b>          | 83%                                              |
| weight                                                | 79%                             | 77%               | 39%                                               | 96%                | 88%                 | <b>90%</b>                                       |
| weight+TTT                                            | 78%                             | <b>80%</b>        | <b>43%</b>                                        | 98%                | <b>95%</b>          | <b>90%</b>                                       |
| RESfM-scratch (supervised)                            | <b>85%</b>                      | 77%               | 37%                                               | <b>100%</b>        | 90%                 | 88%                                              |

so its statistical match at extreme contamination and its clean-OOD wins are not coverage artifacts, and the soft arms register at least as many cameras as supervision under high contamination. These readings, together with the paragraph above, are intended for the camera-ready main text (the Metrics paragraph, Secs. 5.1 and 5.5, and limitation 5).

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Table 15. Extended full shallow mechanism matrix (extends main-paper Table 5; seed 20;  $^{\dagger}$ mean $\pm$ std over seeds 20–22). Translation panel repeats the main table; rotation and  $N_r$  panels: seed-20 means / fractions. Olsson column: means. Best per column in bold.

| Arm                                                   | MD               | 1DSfM                                                   | Strecha            | BMVS              | Olsson       |
|-------------------------------------------------------|------------------|---------------------------------------------------------|--------------------|-------------------|--------------|
| Translation error (mean / median), the primary metric |                  |                                                         |                    |                   |              |
| remove                                                | 0.60/0.19        | 9.1 $\pm$ 2.8/ <b>1.6</b> $\pm$ <b>0.2</b> $^{\dagger}$ | 3.02/1.68          | 0.27/0.19         | 2.81         |
| remove+TTT                                            | 0.49/0.13        | <b>8.7</b> $\pm$ <b>1.1</b> /2.2 $\pm$ 1.1 $^{\dagger}$ | 2.97/1.56          | 0.19/0.07         | 4.07         |
| hybrid                                                | 0.37/0.20        | 16.0 $\pm$ 1.4/6.7 $\pm$ 4.4 $^{\dagger}$               | 2.70/1.39          | 0.10/0.03         | 2.82         |
| hybrid+TTT                                            | 0.39/0.14        | 15.4 $\pm$ 1.1/10.7 $\pm$ 3.5 $^{\dagger}$              | 3.52/2.65          | 0.13/ <b>0.00</b> | 3.19         |
| madweight                                             | 0.42/0.15        | 11.5 $\pm$ 2.3/6.0 $\pm$ 1.4 $^{\dagger}$               | 2.50/0.97          | 0.15/0.004        | <b>2.57</b>  |
| weight                                                | 0.42/0.18        | 15.8 $\pm$ 3.1/10.9 $\pm$ 2.1 $^{\dagger}$              | 2.00/0.03          | <b>0.10</b> /0.04 | 2.65         |
| weight+TTT                                            | 0.39/0.14        | 17.0 $\pm$ 1.3/10.9 $\pm$ 4.1 $^{\dagger}$              | 1.98/ <b>0.005</b> | 0.14/0.01         | 7.74         |
| RESfM (sup.)                                          | <b>0.34/0.07</b> | 11.03/2.93                                              | <b>0.006/0.005</b> | 0.38/0.35         | 8.47         |
| Rotation error ( $^{\circ}$ , means)                  |                  |                                                         |                    |                   |              |
| remove                                                | 4.87             | 5.50                                                    | 12.96              | 19.83             | 12.97        |
| remove+TTT                                            | 4.05             | <b>5.31</b>                                             | 12.81              | 12.52             | 15.60        |
| hybrid                                                | 2.81             | 6.98                                                    | 13.04              | 7.45              | 12.54        |
| hybrid+TTT                                            | 2.99             | 11.32                                                   | 16.62              | 7.80              | 13.76        |
| madweight                                             | 3.49             | 8.31                                                    | 14.62              | 9.60              | 13.57        |
| weight                                                | 3.04             | 8.39                                                    | 7.48               | <b>6.65</b>       | 13.23        |
| weight+TTT                                            | 2.51             | 11.19                                                   | 7.41               | 7.91              | <b>10.49</b> |
| RESfM (sup.)                                          | <b>2.01</b>      | 9.79                                                    | <b>0.02</b>        | 36.10             | 12.34        |
| Registered cameras $N_r$                              |                  |                                                         |                    |                   |              |
| remove                                                | 71%              | 59%                                                     | 96%                | 87%               | 80%          |
| remove+TTT                                            | 68%              | 59%                                                     | 94%                | 90%               | 80%          |
| hybrid                                                | 78%              | <b>82%</b>                                              | 67%                | 90%               | 88%          |
| hybrid+TTT                                            | 77%              | 80%                                                     | 76%                | <b>95%</b>        | 89%          |
| madweight                                             | 80%              | 74%                                                     | 91%                | <b>95%</b>        | 83%          |
| weight                                                | 79%              | 77%                                                     | 96%                | 88%               | <b>90%</b>   |
| weight+TTT                                            | 78%              | 80%                                                     | 98%                | <b>95%</b>        | <b>90%</b>   |
| RESfM (sup.)                                          | <b>85%</b>       | 77%                                                     | <b>100%</b>        | 90%               | 88%          |

Table 16. Extended removal-statistic ablation (extends main-paper Table 6; head reweighting fixed). Translation first, then rotation, then  $N_r$  (1DSfM, seed 20).

| Statistic       | 1DSfM trans                  | med. s20/21/22                       | 1DSfM rot $^{\dagger}$      | $N_r$      | BMVS        |
|-----------------|------------------------------|--------------------------------------|-----------------------------|------------|-------------|
| MAD (madweight) | 11.5 $\pm$ 2.3               | <b>4.4</b> / <b>7.0</b> / <b>6.6</b> | 8.0 $\pm$ 0.8               | <b>74%</b> | 0.15        |
| Huber           | <b>11.5</b> $\pm$ <b>1.6</b> | 3.8 / 12.4 / 5.3                     | <b>7.6</b> $\pm$ <b>0.4</b> | 71%        | <b>0.11</b> |
| STD             | 15.2 $\pm$ 1.2               | 13.3 / 9.0 / 12.9                    | 9.0 $\pm$ 0.2               | <b>74%</b> | 0.15        |

Table 17. Extended per-scene selection (extends main-paper Table 7; mean translation, then rotation, then  $N_r$ ; seed 20). The oracle (italics) selects by ground-truth translation error over five mechanisms; its rotation and  $N_r$  are those of the selected arms.

| Dataset                              | weight       | madwt        | contam     | oracle (GT)  |
|--------------------------------------|--------------|--------------|------------|--------------|
| Translation error (means)            |              |              |            |              |
| MegaDepth                            | 0.423        | <b>0.422</b> | 0.422      | <i>0.116</i> |
| 1DSfM                                | 12.79        | <b>12.39</b> | 12.39      | <i>5.78</i>  |
| Strecha                              | <b>2.00</b>  | 2.50         | 2.00       | <i>1.98</i>  |
| BlendedMVS                           | <b>0.100</b> | 0.146        | 0.100      | <i>0.083</i> |
| Rotation error ( $^{\circ}$ , means) |              |              |            |              |
| MegaDepth                            | <b>3.04</b>  | 3.49         | 3.49       | <i>1.22</i>  |
| 1DSfM                                | 8.39         | <b>8.31</b>  | 8.31       | <i>5.50</i>  |
| Strecha                              | <b>7.48</b>  | 14.62        | 7.48       | <i>7.41</i>  |
| BlendedMVS                           | <b>6.65</b>  | 9.60         | 6.65       | <i>5.28</i>  |
| Registered cameras $N_r$             |              |              |            |              |
| MegaDepth                            | 79%          | <b>80%</b>   | <b>80%</b> | <i>75%</i>   |
| 1DSfM                                | <b>77%</b>   | 74%          | 74%        | <i>66%</i>   |
| Strecha                              | <b>96%</b>   | 91%          | <b>96%</b> | <i>98%</i>   |
| BlendedMVS                           | 88%          | <b>95%</b>   | 88%        | <i>90%</i>   |